AWAIS MANZOOR

1. Project Title

WhatsApp Customer Churn Prediction System using Machine Learning

2. Information

Name: Awais Manzoor (SP24-BBD-026)
Technology Stack: Python, Flask, Twilio, ngrok, Scikit-learn
Model Used: Logistic Regression
Dataset: Customer Churn Modelling Dataset

3. Project Objective

The objective of this project is to build a WhatsApp-based Machine Learning Prediction System that predicts whether a customer will churn or stay in the bank.

The system accepts customer feature values from WhatsApp in comma-separated numeric format, preprocesses the data, applies the trained machine learning model, and sends the prediction result back to the user through WhatsApp automatically.

This project demonstrates the integration of:

- Machine Learning
- Flask Web Application
- Twilio WhatsApp Sandbox
- ngrok Public Tunnel

4. Requirements

The assignment requirements were:

- Select one classification model from the lab mid project
- Build a Flask webhook
- Integrate Twilio WhatsApp Sandbox
- Use ngrok for public access
- Accept numeric input from WhatsApp
- Predict class labels using trained ML model

- Return prediction confidence
- Validate incorrect inputs safely

5. Model Selected

From the lab mid project, the following model was selected:

Model	Type
Logistic Regression	Binary Classification

6. Dataset Information

Dataset used:

Customer Churn Modelling Dataset

Target Column:

Column	Meaning
Exited	0 = Customer Stayed
Exited	1 = Customer Churned

7. Project Folder Structure

```
Assignment_Lab_watsapp/
│
├── app.py
├── preprocessing.py
├── train_and_save.py
├── requirements.txt
├── README.md
│
├── data/
│   └── Churn_Modelling.csv
│
├── artifacts/
│   ├── model.pkl
│   └── scaler.pkl
│
└── demo/
    └── demo_video.mp4
```

8. Required Libraries

Create file:

requirements.txt

```
flask==3.0.0
numpy==1.26.0
pandas==2.2.0
scikit-learn==1.4.0
joblib==1.3.2
twilio==9.0.0
```

Install libraries:

```
pip install -r requirements.txt
```

9. Training the Machine Learning Model

Create file:

- **train_and_save.py**

  Run the training file:

```
python train_and_save.py
```

10. Create Preprocessing File

Create file:

- preprocessing.py

11. Create Flask WhatsApp Bot

Create file:

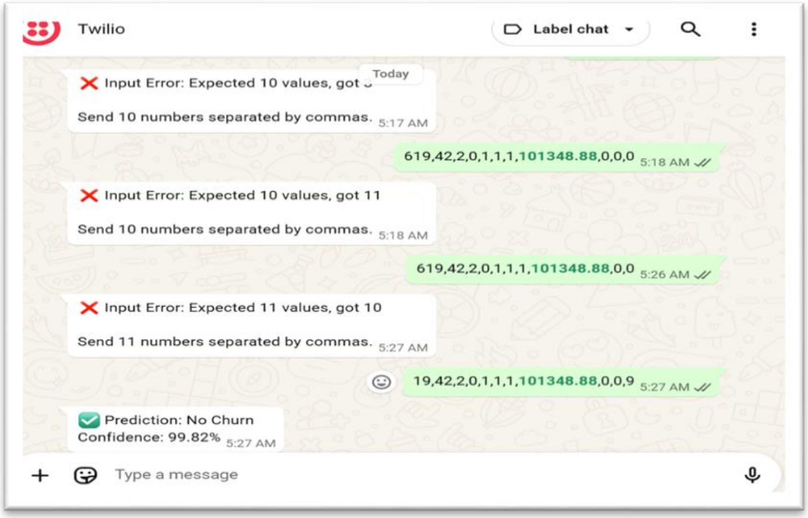
- app.py
- Run Flask Server (python app.py)
- Install and Run ngrok (Start ngrok)

12. Setup Twilio WhatsApp Sandbox

Steps:

- Create a Twilio account
- Open WhatsApp Sandbox
- Join the sandbox from your WhatsApp
- Paste webhook URL
- Method:POST AND SAVE

13. Testing on WhatsApp



14.Complete System Flow

```
WhatsApp User
       ↓
Twilio Sandbox
       ↓
ngrok Public URL
       ↓
Flask Webhook
       ↓
Preprocessing
       ↓
Scaler
       ↓
ML Model
       ↓
Prediction
       ↓
Reply to WhatsApp
```

Conclusion

This project successfully demonstrates how Machine Learning models can be integrated with ***WhatsApp using Flask, Twilio, and ngrok*****.** The system accepts numeric input values from users, preprocesses the data, predicts the class label using a trained ML model, and sends the result back instantly through WhatsApp.